
Modeling Kelp Forest Resistance to Extreme Weather Conditions

Lakshmi Sahiti Vallabhaneni

MATH 142 Mini Project #1 - July 13, 2026

Introduction

Background

Giant kelp (*Macrocystis pyrifera*) grows in dense underwater forests along temperate coastlines. A single kelp plant grows in stages: young sporophytes develop below the surface, and once large enough, they extend upwards to join the floating canopy which receives direct sunlight (Schiel and Foster, 2015). These two layers respond differently to ocean conditions and extreme events such as storms and heat waves. The canopy is directly exposed to wave action, so storms can quickly strip away a lot of the surface growth. Cavanaugh et al. (2011) found that winter canopy loss in the Santa Barbara Channel was strongly related to maximum wave height.

In addition, heatwaves lead to an increase in the temperature of the water, and warm surface water holds fewer nutrients, which negatively affects the growth of giant kelp. Cavanaugh et al. (2019) found kelp's resistance to the 2014 to 2016 marine heatwave was best predicted by peak sea surface temperature, and Bunting et al. (2024) showed growth rates drop sharply above a temperature threshold. Because the two layers respond differently to these stimuli, I aim to create a model that tracks the effects separately while also considering the interactions within the species that may lead to population change.

Problem Statement

This paper builds a mathematical model of a kelp population composed of two groups: the mature canopy at the surface and the young, developing kelp beneath it. I aim to use this model to examine how the size of the two subpopulations changes over time as it is exposed to storms, heatwaves, and other extreme sea weather events: does it grow, stabilize, collapse, or oscillate?

To clarify prompt language, I assume "irregular weather cycles" refers to storms and heatwaves, the two extreme weather events considered in my model. As implied by the term "irregular", they have no fixed schedule or severity (Arafeh-Dalmau et al., 2019); rather, at each time step there is a fixed probability of some weather event happening, in which case the severity is drawn from a range of possible values. Further, we are told to consider "internal interactions" and "environmental stress", so I have chosen this to mean that we only build our model based on extreme weather events and localized births, deaths, and reproduction. More specifically, I have ruled out external spore recruitment from other systems of giant kelp, as this would be considered an external interaction that is not an environmental stressor.

Assumptions

- (1) *Discrete time*: The population is tracked at fixed intervals of 1 month (chosen for compatibility with survey data and typical weather event duration).
- (2) *Two life stages*: Every specimen of Giant Kelp is classified as either "Canopy" or "Below-Surface".
- (3) *Independence of weather-event case from normal case*: Prolonged weather events do not affect the interactions within the kelp beyond the event itself (i.e, kelp exhibits consistent growth/decay in calm conditions).
- (4) *No consumer species or other factors*: There exist no animals that consume the kelp in this ecosystem, and there are no factors other than weather events and kelp itself that affect density.
- (5) *Weather events across seasons and years*: The weather events affect the kelp forests uniformly across all years, seasons and times of the month.
- (6) *Constraint*: Internal dynamics matrix M under calm conditions has $\text{Tr}(M) = 1.175$.

Modeling Approach

I will be building a discrete recursive model given by:
$$\begin{cases} X_t = \begin{pmatrix} C_t \\ B_t \end{pmatrix} \\ X_{t+1} = Mx_t + w_t \end{cases}$$

where M is the 2×2 matrix representing internal interactions under calm conditions, and w_t is a time varying term representing the effect of weather events.

Model Derivation

Internal Dynamics Matrix

Under calm conditions, next month's canopy is the surviving fraction of this month's canopy plus the fraction of below-surface kelp that matures into canopy, and next month's below-surface biomass is the surviving fraction of this month's below-surface population minus baseline shading from the canopy above it:

$$C_{t+1} = m_{11}C_t + m_{12}B_t, \quad B_{t+1} = m_{21}C_t + m_{22}B_t, \quad x_{t+1} = Mx_t.$$

Fronds live 4 to 9 months and canopy turns over about 6 to 7 times per year (Schiel and Foster, 2015). Field frond half-lives from the Falkland Islands refine this further: van Tussenbroek (1989) reports a half-life of 2 to 3 months for fronds in a shallow, wave-exposed zone versus 5 to 6 months in a deeper, sheltered bed, a contrast that parallels the exposed canopy versus sheltered below-surface split used here. Converting each half-life h to a monthly survival fraction via $s = 0.5^{1/h}$ gives $s \approx 0.758$ (shallow) and $s \approx 0.882$ (deep), a ratio of about 0.86:1. Splitting the fixed trace $\text{Tr}(M) = 1.175$ in this ratio gives $m_{11} = 0.543$ and $m_{22} = 0.632$. Maturation is set using a tracked frond growth trajectory in van Tussenbroek (1989): a frond starting at 10 nodes in January reached the canopy-forming threshold of about 30 to 31 nodes by April, a maturation delay of

about 3 months. Treating this delay as the mean time to maturation under a constant monthly rate gives $m_{12} \approx 1/3 \approx 0.33$. I set baseline shading to $m_{21} = -0.05$, since canopy-removal experiments show understory growth increases once canopy is thinned, confirming suppression exists (Schiel and Foster, 2015). This gives

$$M = \begin{pmatrix} 0.543 & 0.33 \\ -0.05 & 0.632 \end{pmatrix}, \quad \text{Tr}(M) = 1.175.$$

Weather Term

Weather is added as its own term w_t :

$$X_{t+1} = MX_t + w_t.$$

$w_t = \begin{pmatrix} w_t^C \\ w_t^B \end{pmatrix}$ is the extra change to canopy and below-surface biomass caused by that month's weather, on top of what M already predicts.

Each month has two severity values: $S_t^{\text{storm}} \in [0, 1]$ and $S_t^{\text{heat}} \in [0, 1]$. A value of 0 means no event that month; a value of 1 means the most severe event considered. These are drawn randomly each month.

A storm removes canopy directly, but the light it frees up helps the below-surface layer. A heatwave weakens canopy, but helps below-surface growth by cooling and easing competition below:

$$w_t^C = -a_C S_t^{\text{storm}} C_t - h_C S_t^{\text{heat}} C_t, \quad w_t^B = a_B S_t^{\text{storm}} C_t - h_B S_t^{\text{heat}} B_t.$$

Each coefficient is named by event type (a for storm, h for heat) and affected layer (C for canopy, B for below-surface). Each is the largest fraction of biomass a fully severe event could move in one month:

- $a_C = 0.5$ (storm loss to canopy): Cavanaugh et al. (2011) found canopy loss climbing sharply with wave height, reaching 70 to 90 percent in the most extreme storms observed.
- $h_C = 0.1$ (heatwave loss to canopy): the same study found sea surface temperature explained canopy recovery weakly ($r^2 = 0.25$) compared to how strongly wave height explained canopy loss ($r^2 = 0.50$), so I set h_C well below a_C .
- $a_B = 0.15$ (storm gain to below-surface): canopy removal reliably increases understory growth (Schiel and Foster, 2015), but light is only one factor in that growth, so I set a_B below a_C .
- $h_B = 0.4$ (heatwave loss to below-surface): Bunting et al. (2024) found below-surface growth dropped more than 30 percent under moderate simulated heatwaves, with worse effects at higher severity.

$$w_t = \begin{pmatrix} -0.5S_t^{\text{storm}}C_t - 0.1S_t^{\text{heat}}C_t \\ 0.15S_t^{\text{storm}}C_t - 0.4S_t^{\text{heat}}B_t \end{pmatrix}, \quad X_{t+1} = MX_t + \begin{pmatrix} -0.5S_t^{\text{storm}}C_t - 0.1S_t^{\text{heat}}C_t \\ 0.15S_t^{\text{storm}}C_t - 0.4S_t^{\text{heat}}B_t \end{pmatrix}$$

Stability Criterion

$$\therefore X_{t+1} = \begin{pmatrix} 0.543 & 0.33 \\ -0.05 & 0.632 \end{pmatrix} X_t + \begin{pmatrix} -0.5S_t^{\text{storm}}C_t - 0.1S_t^{\text{heat}}C_t \\ 0.15S_t^{\text{storm}}C_t - 0.4S_t^{\text{heat}}B_t \end{pmatrix}$$

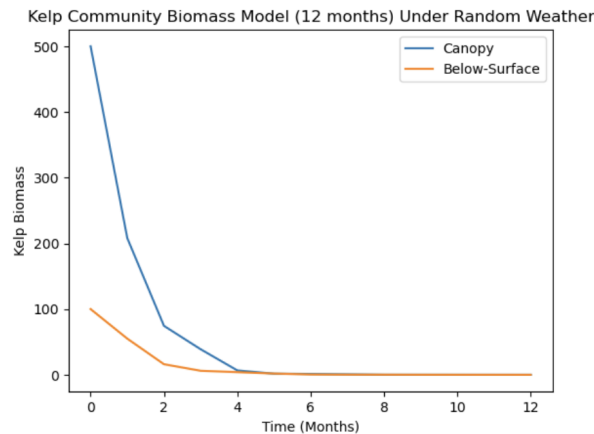
Long-run behavior is governed by the eigenvalues of M :

$$\lambda = \frac{\text{Tr}(M) \pm \sqrt{\text{Tr}(M)^2 - 4 \det(M)}}{2}, \quad \det(M) \approx 0.360.$$

Here $\text{Tr}(M)^2 - 4 \det(M) \approx -0.058 < 0$, so the eigenvalues are complex: $\lambda \approx 0.588 \pm 0.121i$, with magnitude $|\lambda| = \sqrt{\det(M)} \approx 0.600$. Since this magnitude is below 1, the population declines slowly with no weather at all. The eigenvalues being complex means this decline is not smooth: canopy and below-surface biomass rotate between being canopy-heavy and below-surface-heavy from month to month, because the two layers persist and mature at different rates (van Tussenbroek, 1989). The decline itself comes from Assumption 4, which treats the bed as closed with no outside spore input, so the model captures only internal redistribution and weather stress.

Answering the Prompt

Under calm conditions alone, the model already declines slowly, with mild oscillation between canopy and below-surface biomass, since M 's eigenvalue magnitude is below 1. Adding weather does not change this basic outcome: storms and heatwaves reduce the eigenvalue magnitude of M_t further in the months they occur, so the population never grows or stabilizes under this model, only declines faster or slower depending on how often and how severely weather hits. This means that as extreme weather becomes more frequent and severe under future climate conditions, the community is predicted to decline more quickly, while a community facing only rare, mild events declines close to the slow calm-condition rate.



Model Assessment

Strengths

M captures internal interactions, canopy effects, maturation, and shading, while w_t captures weather stress separately, so the two effects the prompt names stay distinct.

Because w_t scales with current biomass, the system can still be written as $x_{t+1} = M_t x_t$, so eigenvalues give a clean, general answer to whether the community grows, declines, or oscillates, without needing a full simulation. The model also captures a real mechanism: storms remove canopy but free light for below-surface growth, matching reported increases in understory growth after canopy loss (Schiel and Foster, 2015).

Weaknesses

If my model reflected the real world, Giant Kelp would be extinct, which is not the case. Simplifying assumption 4 treats the community as closed, with no outside spore input. This makes the calm-condition eigenvalue magnitude fall below 1, so the model declines even with no weather at all, and cannot show real stabilization or recovery. The model is also linear, so it has no upper limit on population size if conditions were mild enough to grow, i.e., my model does not consider overall limits on nutrients and space. Weather is assumed equally likely every month, which does not match real seasonal storm and heatwave patterns (Cavanaugh et al., 2019), and grazers like sea urchins are left out entirely (Man et al., 2025).

Conclusion

This paper modeled a kelp community as two linked groups, canopy and below-surface biomass, using a matrix M for internal interactions and a separate term w_t for storm and heatwave stress. Calibrated using field data from the literature, the model showed that even under calm conditions the population declines slowly, with mild oscillation between canopy-heavy and below-surface-heavy states, since canopy and below-surface kelp persist and mature at different rates. Adding weather does not change this basic outcome: storms and heatwaves only make the decline faster and the oscillation stronger, never causing growth or stabilization. This means that under the assumptions used here, a more frequent and severe future weather regime is expected to shorten the time to a critically small population, while rare, mild events allow the community to decline much more slowly.

References

- Arafeh-Dalmau, N., Montaña-Moctezuma, G., Martínez, J.A., Beas-Luna, R., Schoeman, D.S., and Torres-Moye, G. (2019). Extreme Marine Heatwaves Alter Kelp Forest Community Near Its Equatorward Distribution Limit. *Frontiers in Marine Science*, 6:499. doi: 10.3389/fmars.2019.00499
- Bunting, I., Kok, Y.Y., Krieger, E.C., Bury, S.J., D'Archino, R., and Cornwall, C.E. (2024). Marine heatwave intensity and duration negatively affect growth in young sporophytes of the giant kelp *Macrocystis pyrifera*. *Frontiers in Marine Science*, 11:1423595. doi: 10.3389/fmars.2024.1423595
- Cavanaugh, K.C., Siegel, D.A., Reed, D.C., and Dennison, P.E. (2011). Environmental controls of giant-kelp biomass in the Santa Barbara Channel, California. *Marine Ecology Progress Series*, 429, 1-17. doi: 10.3354/meps09141
- Cavanaugh, K.C., Reed, D.C., Bell, T.W., Castorani, M.C.N., and Beas-Luna, R. (2019). Spatial Variability in the Resistance and Resilience of Giant Kelp in Southern and Baja California to a Multiyear Heatwave. *Frontiers in Marine Science*, 6:413. doi: 10.3389/fmars.2019.00413
- Man, L., Barbosa, R.V., Reshitnyk, L.Y., Gendall, L., Wachmann, A., Dedeluk, N., Kim, U., Neufeld, C.J., and Costa, M. (2025). Canopy-forming kelp forests persist in the dynamic subregion of the Broughton Archipelago, British Columbia, Canada. *Frontiers in Marine Science*, 12:1537498. doi: 10.3389/fmars.2025.1537498
- Schiel, D.R. and Foster, M.S. (2015). *The Biology and Ecology of Giant Kelp Forests*. University of California Press, Oakland, CA.
- van Tussenbroek, B.I. (1989). The life-span and survival of fronds of *Macrocystis pyrifera* (Laminariales, Phaeophyta) in the Falkland Islands. *British Phycological Journal*, 24:2, 137-141. doi: 10.1080/00071618900650131

AI Use Disclosure

I used Claude to answer some questions about the prompt, I verified these details independantly and did not incorporate it in any manner in my report.

Claude (Jul 10, 2026 version, Sonnet 4.6,)

Query: what is a trace invariant

Purpose: Output was NOT incorporated in the code, was used for understanding a mathematical term.

Claude (Jul 10, 2026 version, Sonnet 4.6,)

Query: trace invariant meaning in the context of building a mathematical model with stage matrix M

Output: Output was NOT incorporated in the code, was used for understanding a mathematical term.

Besides this, Overleaf's grammar correct AI gave me grammar suggestions as I was typing that I occasionally used, and I used Citation Machine by Purdue OWL to draft my references.

Code

```
[1]: # imports
import numpy as np
import matplotlib.pyplot as plt

[5]: # Set Variables
ncensus = 12
M = np.array([[0.543, 0.33],
              [-0.05, 0.632]])
aC, hC, aB, hB = 0.5, 0.1, 0.15, 0.4
P = np.zeros([ncensus+1, 2])

# Initialize all at same point
P[0] = [500, 100]
for k in range(ncensus):
    # randomly draw storm and heatwave severity for this month
    S_storm = np.random.uniform(0,1)
    S_heat = np.random.uniform(0,1)

    # weather term w_t, added separately from M
    w = np.array([-aC*S_storm*P[k,0] - hC*S_heat*P[k,0],
                  aB*S_storm*P[k,0] - hB*S_heat*P[k,1]])

    P[k+1] = M @ P[k] + w

#Plotting
plt.plot(P)
plt.xlabel('Time (Months)')
plt.ylabel('Kelp Biomass')
plt.legend(['Canopy', 'Below-Surface'])
plt.title('Kelp Community Biomass Model (12 months) Under Random Weather')
```